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# Parathyroid gland function and the set point for PTH release: understanding the available data

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Dynamic tests of parathyroid gland function have been used by several groups of investigators to evaluate parathyroid hormone (PTH) secretion in patients with various disorders of mineral metabolism including chronic renal failure [1–7]. Elsewhere in this issue, Malberti and colleagues describe changes in the pattern of PTH release in patients with secondary hyperparathyroidism after 6 months of intermittent calcitriol therapy and after subtotal parathyroidectomy. Similar results have been reported by others following intermittent calcitriol therapy [1,3,8–10], and such findings have been taken as evidence that 1,25-dihydroxyvitamin D either partially or fully corrects an intrinsic pathophysiological abnormality in the regulation of PTH secretion in secondary hyperparathyroidism. Specifically, calcitriol is thought to lower the set point for calcium-regulated PTH release in this disorder rendering the parathyroids more sensitive to the suppressive effects of calcium on PTH secretion [1,11].

Two recent reports indicate, however, that the regulation of PTH release by calcium does not differ from normal in patients with chronic renal failure [7,12]. In contrast to previous investigations and to the work of Malberti *et al.* [3,10], both studies provided data obtained in normal volunteer subjects for comparison with results from patients with secondary hyperparathyroidism [7,12]. It should be noted that the methods employed by Ramirez *et al.* and by Messa *et al.* differ substantially from those used by others to report that calcitriol lowers the set point for calcium-regulated PTH release and that the set point differs in the various types of renal osteodystrophy [13,14]; these methodological differences warrant further consideration.

Mayer and co-workers originally described the inverse sigmoidal relationship between the concentration of ionized calcium in serum and the serum level of parathyroid hormone (PTH) across the physiological range of ionized calcium values in studies in the calf [15]. Subsequent work by Brown and associates using dispersed parathyroid cells obtained from patients with parathyroid adenomas or parathyroid gland hyperplasia documented a similar relationship between the amount of PTH released *in vitro* and the calcium level in tissue culture media [2,16–19]. Because calcium is the major regulator of PTH release, curve-fitting models that had been applied to the assessment of other physiological functions with non-linear properties such as enzyme kinetics and ligand-receptor interactions were used to examine the regulation of PTH release by calcium *in vitro* [2,20]. These assessments sought to better characterize the components of PTH secretion at the cellular level and to determine whether the control of PTH secretion by calcium differed in various disease states [6]; they also form the basis for *in-vivo* studies of the dynamics of PTH secretion in patients with secondary hyperparathyroidism and other disorders of mineral metabolism.

As depicted by the following equation, the model employed by Brown and colleagues provides estimates of the amount of PTH (Y) released at any concentration of ionized calcium (X) using four parameters derived by direct measurement: A, the maximum PTH value observed at reduced ionized calcium levels; D, the minimum PTH value observed at high concentra-

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tions of ionized calcium; C, the set point, which is defined as the ionized calcium level at which PTH values are midway between A and D; and B, the slope of the sigmoidal curve at the set point (4,17) (see Figure 1).

$$Y=\frac{A-D}{1+)X/C)^{B}}$$

Because there is considerable heterogeneity among parathyroid cell preparations with respect to the amount of PTH released per 10 000 cells in tissue culture, data for *in-vitro* studies are generally normalized by expressing PTH secretion as a percentage of the maximum amount of hormone released at low calcium concentrations [2,20]; this approach permits comparisons among tissue preparations with differing maximal secretory capacities, and it provides information about the secretory behaviour of individual cells because the total number of cells in the preparation is known [2].

Unlike the situation *in vitro*, there is currently no reliable method for determining the size of the parathyroid glands *in vivo*; thus, although PTH release can be normalized for tissue size and cell number for *in-vitro* studies, serum PTH levels obtained *in-vivo* reflect not only parathyroid gland size but also the intrinsic secretory function of individual parathyroid cells. Nevertheless, early studies by Brown and colleagues in patients with primary hyperparathyroidism suggested that estimates of the set point for calcium-regulated PTH release determined *in vivo* by varying serum ionized calcium levels during infusions of sodium EDTA and calcium gluconate corresponded with values measured *in vitro* in parathyroid tissues obtained at surgery from the same patients [19].

As developed originally by Delean and co-workers and applied by Brown *et al.* [2,20], the set point for calcium-regulated PTH release represents the ionized calcium level at which PTH values are midway between the maximum level achieved at low calcium concentrations and the minimum level observed at high calcium concentrations. In their current as well as in their previous work, Malberti *et al.* have employed an analytical model used by Felsenfeld and colleagues to assess PTH secretion in patients undergoing regular haemodialysis [4]. In contrast to the four-parameter model, this approach defines the set point as the serum ionized calcium at which PTH levels are 50% of the maximum value attained during hypocalcaemia (see Figure 1). Because PTH secretion persists at basal levels, even at very high ionized calcium concentrations, PTH values rarely if ever fall to levels approaching 0% of maximum [2,4,12]. Accordingly, the set point obtained using this definition is located below the midpoint of the inverse sigmoidal curve at a correspondingly higher ionized calcium level than estimates obtained using the four-parameter model (see Figure 1).

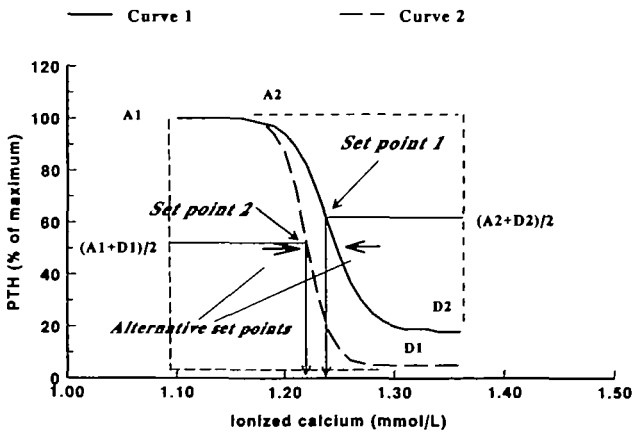


Fig. 1. Comparison of two sigmoidal curves with different configurations. Set point values obtained using the four-parameter model using the equation (A+D)/2 are depicted as set point 1 and 2 respectively. The short horizontal arrows show set point values for each curve using an alternative definition of 50% of maximum PTH; this approach overstates the difference between set point values compared to the four-parameter model.

A comparison of set point values for two sigmoidal curves becomes particularly problematic using this alternative model of PTH secretion if the lowest PTH values achieved during hypercalcaemia differ between curves when expressed as a percentage of maximum values. Under these circumstances, the lower portions of the two curves diverge as ionized calcium levels increase, whereas the highest values for each curve remain fixed at 100% (see Figure 1). The set point for the sigmoidal curve with the higher minimum PTH values will therefore be located above and slightly to the right of the set point for the adjacent curve when determined using the four-parameter model. In contrast to this result, the set point for the uppermost curve is displaced further below the mid-portion of the ionized calcium-PTH relationship in the model of Felsenfeld *et al.* when a PTH value that is 50% of maximum is used to define the set point for both curves. As such, difference between set point values for two asymmetric curves are overestimated with this approach [12].

In presenting data for the sigmoidal curve, PTH values are plotted on the ordinate against corresponding ionized calcium levels on the abscissa. For *in vivo* studies, blood samples are collected at predetermined intervals during the induction of hypocalcaemia or hypercalcaemia without knowledge of the prevailing ionized calcium level at the time the sample is obtained. In most studies PTH levels for curve analyses have been plotted at regular intervals of ionized calcium, such as increments of 0.05 or 0.10 mmol/l, indicating that the mean values represent a range of ionized calcium levels both above and below selected points along the X axis [3,10,13,14]. Results plotted in this manner have variance, therefore, along the X axis for ionized calcium as well as along the Y axis for PTH, and the use of standard deviation or standard error bars is required to indicate the variability of ionized calcium at each level of PTH. Such information is necessary not only to determine the degree of overlap for ionized calcium values at various PTH levels for

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two adjacent sigmoidal curves levels but also to provide confidence intervals for estimates of the set point for each sigmoidal curve.

To calculate the slope the sigmoidal curve, Malberti *et al.* and Felsenfeld *et al.* have used a linear connection between two points corresponding to values that are 95% of the maximum PTH level achieved during hypocalcemia and 5% above the minimum PTH level observed during hypercalcemia [4,13]. Slope estimates obtained in this manner differ intrinsically from those determined using the four-parameter model. Linear projections using values near the upper and lower limits of observed values for ionized calcium and PTH yield lower slope estimates than with the four-parameter model because the true slope of the sigmoidal curve is greatest at or near its mid-portion. Small errors in the measurement of PTH or ionized calcium may also produce substantial variation in slope estimates using this method.

As a general rule, estimates of slope and the confidence interval for predicted values for points along a curve, such as the set point, are rendered more precise by increasing the number of measurements along the mid-portion of the curve; this issue is particularly important in enzyme kinetics studies for estimates of Km, or half maximal enzyme activity, and in ligand-binding studies for estimates of Kd, or receptor binding affinity [21]. Unfortunately, the methods used to vary serum ionized calcium levels in nearly all *in vivo* studies of patients with secondary hyperparathyroidism or other forms of renal osteodystrophy induce rapid and substantial changes in serum PTH levels with only small changes in ionized calcium; thus, realtively few data points are located along the mid-portion of the ionized calcium–PTH curve, undermining the precision of estimates within this range [1,12–14, 22].

To date, there have been no direct comparisons between results obtained using any of several alternative methods to evaluate the dynamics of PTH release *in vivo* and those determined using the classic four-parameter model. In contrast to work done by Brown *et al.*, evidence has yet to be presented that *in-vivo* estimates of the set point using alternative modes of data analysis correspond to findings obtained in parathyroid tissues studied *in vitro*. Until such information becomes available the physiological meaning of results obtained from *in-vivo* studies of parathyroid gland function using techniques that differ from the four-parameter model remains uncertain.

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## Editor's note

Please see also the Original Article by Malberti (pp. 81–87 in this issue).

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